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ORIGINAL ARTICLE

A signal quality assessment–based ECG waveform delineation method used for wearable monitoring systems

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Abstract

Identifying transient and nonpersistent abnormal electrocardiogram (ECG) waveforms by continuously monitoring the high-risk populations is of great importance for the diagnosis, treatment, and prevention of cardiovascular diseases. In recent years, fabric electrodes have been widely used in wearable devices because of their non-irritating properties and better comfort than traditional AgCl electrodes. However, the motion noise caused by the relative movement between the fabric electrodes and skin affects the quality of ECGs and reduces the accuracy of diagnosis. Therefore, delineating the ECG waveforms that are recorded from wearable devices with varying levels of noise is still challenging. In this study, a signal quality assessment (SQA)–based ECG waveform delineation method that is used for wearable systems was developed. The ECG signal was first preprocessed by a bandpass filter. Five indices, including the multiscale nonlinear amplitude statistical distribution ($adSQI_1$, $adSQI_2$), the proportion of energy-related to T wave (ptSQI), and heart rates computed from R waves and T waves (rHR and tHR, respectively), were then calculated from the preprocessed ECG signal. The signals were classified as good, acceptable, and unacceptable ECGs by combining these indices through the use of a neural network. Subsequently, the R waves or/and T waves were identified for the corresponding feature interpretations based on the SQA results. ECGs that were recorded from the chest belts from 29 volunteers at different activity statuses were divided into 4-s segments. A total of 7133 manually labeled segments were used to derive (4985 segments) and validate (2148 segments) the algorithm. The $adSQI_1$, $adSQI_2$, tHR, and rHR characteristics were significantly different among the good, acceptable, and unacceptable ECGs. The ptSQI value was considerably higher in the good ECGs than in the acceptable and unacceptable ECGs. The ECG segments of different quality levels were classified with an accuracy of 96.74% by using the proposed SQA method. The R waves and T waves were identified with accuracies of 99.95% and 99.57%, respectively, for segments that were classified as acceptable and/or good. The SQA-based ECG waveform delineation method can perform reliable analysis and has the potential to be applied in wearable ECG systems for the early diagnosis and prevention of cardiovascular diseases.

Keywords Electrocardiography · ECG waveform delineation · Signal quality assessment · Signal quality indices

1 Introduction

Cardiovascular disease (CVD) is one of the leading causes of death worldwide, with an estimated 17.9 million deaths annually, according to the latest report from the World Health Organization. Fortunately, it is reported that 90% of the CVDs, such as arrhythmia, ischemia, ventricular hypertrophy, bundle branch block, and myocardial infarction, are preventable [1]. Because an electrocardiogram (ECG) reflects the electrical activity of the heart, the temporal and morphological characteristics of the ECG signal contain abundant physiological and pathological information, which is helpful in the diagnosis and early warning of diseases [2, 3]. For instance, the alternans of the R and T wave amplitude

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has been used for the prediction of ventricular arrhythmia. Alterations in the QTc intervals are associated with sudden cardiac death. Continuous ST segment monitoring has been shown to provide reliable information about the presence of myocardial ischemia in patients with coronary artery disease. Heart rate variability (HRV) is a reliable reflection of the many physiological factors that modulate the normal rhythm of the heart and has a well-established role as a marker of cardiovascular risk [4–6]. However, traditional ECG monitoring devices can be used only in hospitals and require high levels of professional knowledge [7–9]. Additionally, AgCl electrodes used with these devices may irritate the skin, especially when used for long-term monitoring [10].

In recent years, the advances of wearable technologies and digital signal processing technologies have enabled the continuous monitoring of ECGs and human physical activities during daily life. These wearable devices have been utilized not only for the prevention of diseases and maintenance of health but also for patient management and disease management [11–14]. However, various noise types that occupy the frequency band of the ECG signals, such as electromyogram (EMG) interference and motion artifacts, are simultaneously recorded by wearable devices [15–17]. The presence of noise reduces the quality of ECGs, which leads to decreased signal-to-noise ratio (SNR). A previous study has shown that the SNR limits for reliable R-wave detection and full ECG waveform analysis are 5 and 18 dB, respectively [18]. The fluctuations in SNRs decrease the robustness of the ECG fiducial point extraction method and thereby directly impacts clinical decision-makings.

Since most of the clinically useful information in ECG signals needs to be extracted based on accurate and robust ECG waveform delineation, many techniques have been developed to address the problem of a high false alarm rate due to the noise in wearable monitoring systems [19]. One technique is ECG denoising, such as the adaptive Fourier decomposition filter, transform domain singular value decomposition filter, and wavelet-based Wiener filter, which are used to suppress muscle and electrode motion noise [20–22]. Unfortunately, these filtering methods may also remove some useful ECG attributes while filtering out the noise [23]. Another technique is the signal quality assessment (SQA) method, such as the amplitude and slope criteria threshold–based method, cepstrum, i.e., the inverse Fourier transform of the logarithm of spectrum-based, regularity matrix, and multifeature fusion–based methods [24–27]. These methods classify ECGs into acceptable and unacceptable quality signals mainly based on the R wave–related features. Unacceptable signals are discarded and only acceptable signals are analyzed. These SQA methods play a vital role in significantly improving the diagnostic accuracy and reliability of unsupervised ECG analysis systems [28]. However, the disadvantage is that a large number of ECGs corrupted with noise must be discarded when the human motion activity increases. A failure of continuous monitoring will lead to a loss of the transient and nonpersistent abnormal ECG waveforms in high-risk populations, especially for subjects with acute CVDs. Indeed, many studies have suggested the usefulness of HRV analysis in the prediction of sudden cardiac death and the usefulness of T wave–correlated characteristics in the diagnosis of myocardial ischemia [29–31]. Therefore, the identification of transient and nonpersistent abnormal ECG waveforms with varying levels of quality by continuously monitoring high-risk populations is still challenging.

In this study, an SQA-based ECG waveform delineation method that was used for wearable systems was proposed. The method was developed and tested on ECG data that were recorded through a custom designed wearable chest belt under different activity statuses.

2 Materials and methods

2.1 Architecture of wearable chest belt system

The architecture of the wearable chest belt system is shown in Fig. 1. It consists of an ECG acquisition module, transmission module, and display and analysis module. In the ECG acquisition module, three conductive textiles filled with sponge are embedded on a flexible chest belt and served as ECG electrodes. The sensed ECG signal, together with the auxiliary acceleration signals (e.g., x_i, y_i, and z_i), are amplified, filtered, digitized, and stored into an MCU-based data acquisition circuit. The chest belt is placed between the fourth and fifth ribs with the data acquisition circuit put in the precordium. The sample rate is 250 Hz for ECG and 100 Hz for acceleration. The transmission module transmits the digitalized data to terminal devices via Bluetooth. An Android-based smartphone is used as the terminal device to display and analyze the received data. We have validated that waveforms can be communicated in an uninterrupted manner via the Bluetooth at up to 5 m away from the smartphone, at a baud rate of 115,200 by using an ECG simulator (ProSim™ 8, Fluke Biomedical, Cleveland, OH, USA).

2.2 Data collection and annotation

Data were collected from 29 male adult volunteers during physical activities (e.g., resting, walking with reference speeds of 4 and 6 km/h, running with reference speeds of 8 and 12 km/h, and creeping). The study was approved by the Ethics Committee of the Army Military Medical University (ID: AF/SC-08/1.0). All participants agreed to the experimental protocol before enrollment into this study.

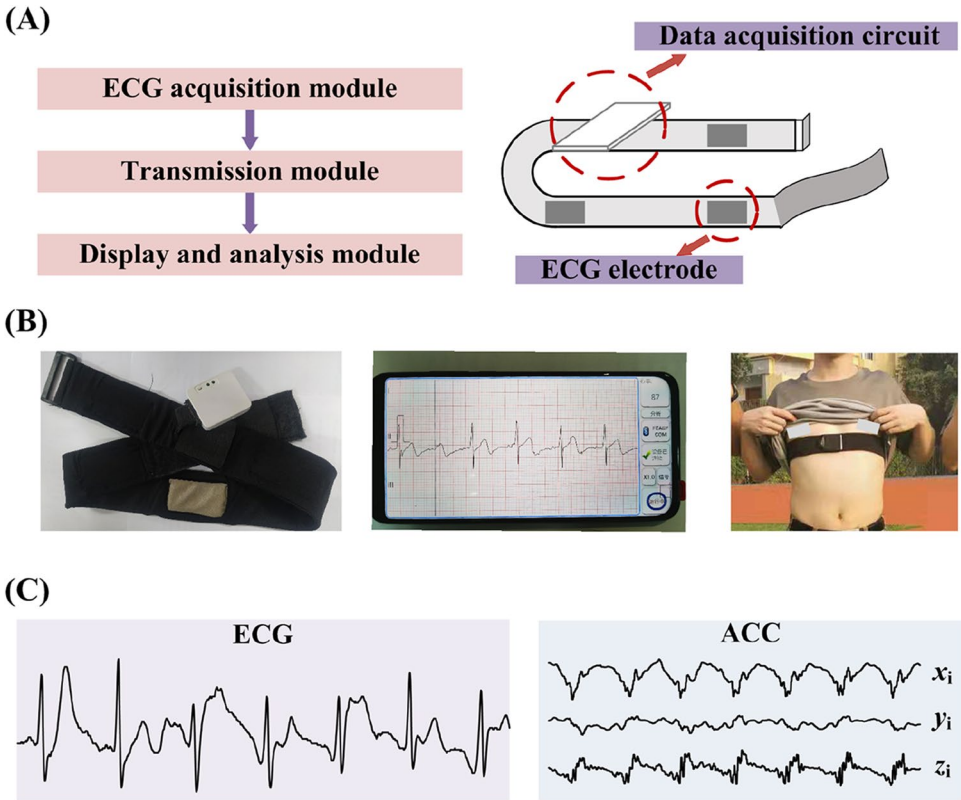

Fig. 1 The architecture of the wearable chest belt system used in this study. **A** Schematic diagram of the chest belt; **B** prototype of the chest belt; **C** illustrations of the ECG and acceleration (ACC) signals

The collected data were then cut into 4-s nonoverlapping segments. A 4-s segment was used because the previous study has found that 4 s was the most appropriate windowing length for ECG waveform analysis [32]. Since overlapping windows will cause more segments to be labeled as good/acceptable than unacceptable and then result in an overestimation of the classification accuracy, we used nonoverlapping windows in this work [33]. Each segment was annotated as good, acceptable, or unacceptable ECG by two experienced medical doctors according to the morphology of the waveform. A detailed description of each of the quality levels and the annotation criteria are summarized in Table 1. If there was a disagreement between the two doctors, another expert was involved, and a consensus was obtained after discussion. Figure 2 shows an example of the ECG segments that were annotated as different signal quality levels. The signal magnitude area (SMA) of the acceleration signal in each segment was calculated using the following equation.

$$SMA = \frac{1}{w}\left(\sum_{i=1}^{w}|x_i| + \sum_{i=1}^{w}|x_i| + \sum_{i=1}^{w}|z_i|\right) \quad (1)$$

2.3 Framework of the proposed method

A flowchart of the SQA-based ECG waveform delineation method is shown in Fig. 3. The ECG segment is first preprocessed by a bandpass filter. The signal quality of the preprocessed ECG is then assessed and presented as a signal quality grade (SQG). If SQG = 0, the segment is classified as unacceptable ECG and discarded. If SQG = 1, the segment is classified as an acceptable ECG. The R waves are detected, and the HRV-related parameters are analyzed. Otherwise, the segment is classified as good ECG, and both the R waves and T waves are detected. The

Table 1 ECG quality levels and criteria for annotation

Quality level	Criterion for annotation
Good	ECGs have clear R waves and T waves and relatively steady line between beats
Acceptable	The signal is subject to moderate interference, all R waves can be recognized, but T waves are less clear to identify
Unacceptable	The signal is heavily interfered, the R waves are partially or completely covered by other interferences, not all R waves can be recognized

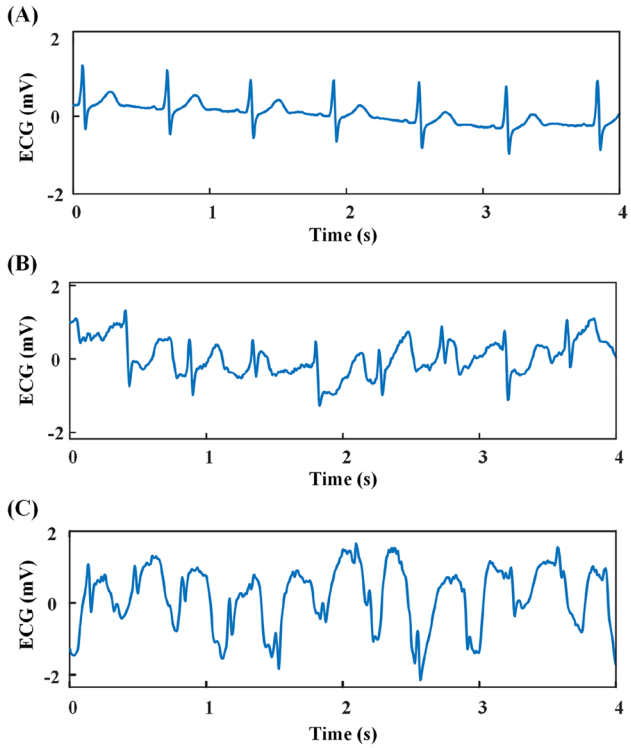

Fig. 2 Examples of ECG signals that were acquired from wearable chest belts with different visually labeled quality levels. **A** Segment labeled as a good ECG; **B** segment labeled as an acceptable ECG; **C** segment labeled as an unacceptable ECG

HRV-related parameters together with the morphological-related parameters are analyzed.

1. Preprocessing
   Baseline wander is excluded from the raw ECG by using an elliptical high-pass filter, which has precisely zero-phase distortion with a cutoff frequency of 0.50 Hz. This frequency is selected because a previous study reported that a 0.5-Hz high-pass filter could effectively suppress the low-frequency noise and motion artifacts that significantly overlap with the signal spectrum of the ECG [28]. High-frequency signals that are outside of the band of interest are eliminated by an elliptical Butterworth filter with a cutoff frequency of 75 Hz, which is an all-zero filter.
2. Signal quality assessment
   Five features that were extracted from the preprocessed ECG segment were used as the inputs to a back-propagation (BP) neural network. The signals were then classified by using one of the three quality levels based on the output of the neural network.
3. Features extraction

Indices that are related to the amplitude distribution adSQI$_1$ and adSQI$_2$ Since EMG interference and motion artifact are mixed with the ECG signal and changed the signal amplitude distribution during physical activity, these indices were extracted in our work. The QRS spikes in the preprocessed signal are extracted by signal conversion using the following equation:

$$y[n] = \left| 2x[n] - x\left[n - \left\lfloor \frac{af_s}{2} \right\rfloor\right] - x\left[n + \left\lfloor \frac{af_s}{2} \right\rfloor\right]\right| \tag{2}$$

where $x[n]$ is the preprocessed signal, f_s is the sampling rate of the ECG, and $\lfloor\ \rfloor$ represents rounding-down. The parameter α is a coefficient that is related to the width of the QRS wave and is optimized to 0.06 s in this work. The $y[n]$ values are sorted in ascending amplitude order, and the points that exceed the amplitude threshold AT are then removed. The AT is obtained by the interquartile range (IQR) and the third quartile (Q3) of the sorted signals, as shown in the following equation:

$$\mathrm{AT} = 3 \times \mathrm{IQR} + \mathrm{Q3} \tag{3}$$

The remaining points are divided into 10 intervals based on their amplitudes, and the total number of data points in each interval, n_i (i = 1,…, 10), in the histogram is then counted.

The indices, adSQI$_1$ and adSQI$_2$, are calculated as:

$$\mathrm{adSQI}_1 = \frac{n_1}{(n_4 + n_5)} \tag{4}$$

$$\mathrm{adSQI}_2 = \frac{n_1}{n_2} \tag{5}$$

The proportion of the main peak spectrum that is associated with T-wave ptSQI and calculated heart rates based on the T wave tHR These indices were analyzed because we observed that the T wave characteristics can be used to indicate the degree of contamination of the ECG signal. Low-pass filtering is performed on the preprocessed signal ($x[n]$) to obtain an ECG with a frequency below 2.5 Hz ($b[n]$). A window with 45 sampling points in length is used to calculate the peak-to-peak value (pp$v$) with a sliding step of one sample point to generate a continuous pp$v$ signal. Then, the pp$v$ signal is transformed into a signal that is similar to a sinusoidal signal ($s[n]$) by smoothing filtering. The main frequency (f_{max}) and peak width ($wide$) of signal $s[n]$ are calculated by use of the fast Fourier transform (FFT).

$$f_{max} = max(P(f)) \tag{6}$$

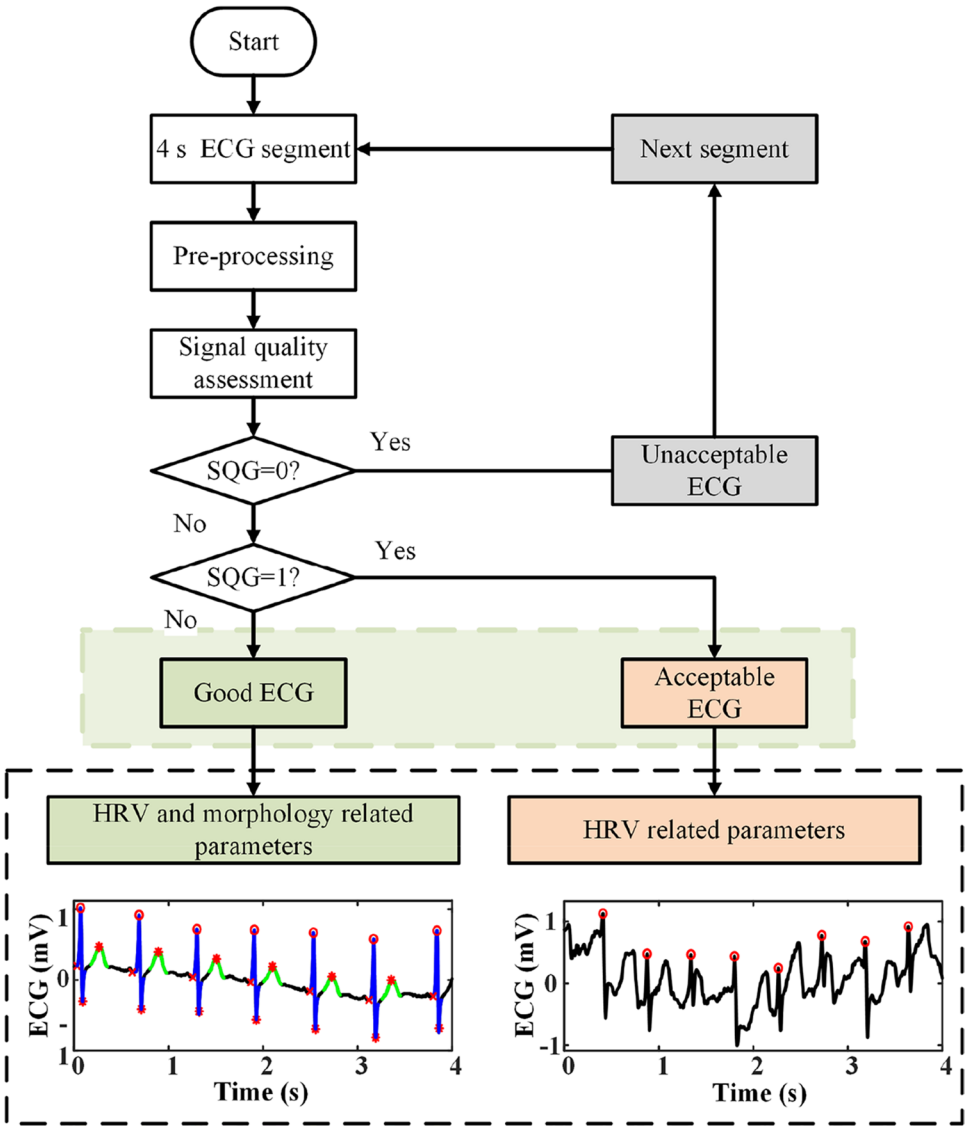

Fig. 3 The signal quality assessment–based ECG waveform delineation method. SQG, signal quality grade

$$wide = \left\lfloor \frac{50}{W/f_s} \right\rfloor \times \frac{f_s}{N} \tag{7}$$

$P(f)$ is the amplitude spectrum, N is set to 2^{14}, and W is the length of $s[n]$. ptSQI is the proportion of the main peak spectrum, as shown in the following equation:

$$ptSQI = \frac{\int_{f_{max}-wide}^{f_{max}+wide} P(f)df}{\int_0^{N/2} P(f)df} \tag{8}$$

The heart rate is based on the T wave and is calculated by:

$$tHR = 60 \times f_{max} \tag{9}$$

Calculated heart rates based on the R wave rHR This index is calculated because the QRS complex is the most striking waveform within the ECG. High-pass filtering is performed on the preprocessed signal to obtain an ECG with a frequency above 5.50 Hz. A window with a sampling interval of 0.08 s is first used to calculate the pp$v$ signal. Then, the window width is adjusted to dilate the peak. Finally, after smoothing, N point FFT is performed to extract the main peak and the rHR is obtained in the frequency domain.

4. Neural network for classifying ECG signal quality

A three-layer BP neural network is used to train and test the algorithm because previous studies have shown that neural networks provide the best classification accuracies compared with other machine learning methods for determining the acceptability of ECGs [34, 35]. The input layer of the designed network has 5 neurons that represent the 5 extracted features obtained in feature extraction section. The output layer has 3 neurons, which

represent 3 different signal quality levels. The optimal number of nodes in the hidden layer was determined to be 13, which was based on empirical formulas and experiments. The hyperbolic tangent sigmoid transfer function is used as the activation function in the input, hidden, and output layers. Each layer consists of the weights, a bias vector, transfer function, and output vector.

5. R wave and T wave detection

The power-based JQRS algorithm was used to detect the R waves for the acceptable and good ECGs [36, 37]. This algorithm consists of a window-based peak energy detector. The original bandpass filter was replaced with a QRS-matched filter (Mexican hat) and used an additional heuristic that ensured that no detections were made during flat lines. A search back procedure was allowed for cases of suspected missed beats [38]. The T waves were then detected by singularity and were corrected by an RR interval–based T wave peak correction strategy [39].

2.4 Performance evaluation

The sensitivity (Sen) and accuracy (Acc) are used to evaluate the performance of the algorithm. The sensitivity is defined as:

$$Sen = \frac{TP}{TP + FN} \times 100\% \tag{10}$$

where *TP* represents true positive values and *FN* represents false negative values. The accuracy is defined as:

$$Acc = \frac{TP + TN}{N} \times 100\% \tag{11}$$

N is the number of segments and *TN* represents true negative values. The consistency between the annotation and estimated signal qualities was analyzed by the use of kappa statistics.

2.5 Statistical analysis

All features of the training dataset were examined for normal distribution using the Kolmogorov–Smirnov Test. If the numerical attributes were normally distributed, the indices are presented as the mean ± SD and were compared by analysis of variance (ANOVA). Otherwise, the median values with their corresponding first and third quartiles (Q1–Q3) were reported, and the *Mann–Whitney U* test was used. The sensitivity and accuracy were compared by using the chi-square test. A $p < 0.05$ was considered to be statistically significant.

3 Results

The average ECG recording time was 16.43 ± 2.93 min, and a total of 7133 segments were obtained. Three thousand one hundred thirty-nine segments (44.01%) were labeled as good, 2861 segments (40.11%) were labeled as acceptable, and 1133 (15.88%) were labeled as unacceptable. The SMAs were 0.73 (0.61, 0.93) for the unacceptable ECGs, which were significantly greater than those for the good (0.15 (0.05, 0.18)) and acceptable (0.20 (0.16, 0.35)) ECGs ($p < 0.01$).

Each quality level was randomized at a ratio of 7:3 to generate a training dataset and testing dataset. A total of 4985 segments were included in the training dataset, which included 2190 good ECGs, 2002 acceptable ECGs, and 793 unacceptable ECGs. The remaining 2148 segments were included in the testing dataset, which included 949 good ECGs, 859 acceptable ECGs, and 340 unacceptable ECGs.

3.1 Differences of the indices at different quality levels

Figure 4 illustrates the differences in the amplitude distribution–related indices, $adSQI_1$ and $adSQI_2$, which were derived from ECG segments at different quality levels. For the good ECGs, both the $adSQI_1$ and $adSQI_2$ values were high due to the uniformly distributed and concentrated peaks. For the acceptable ECG, $adSQI_1$ had a medium value, while $adSQI_2$ had a low value because of the unevenly scattered amplitude of the peaks. For the unacceptable ECGs, both $adSQI_1$ and $adSQI_2$ had low values since the amplitudes of the peaks were unevenly scattered and also were not concentrated.

Figure 5 indicates the differences in ptSQI and tHR that were calculated from the ECG segments with different quality levels. For the good ECGs, the characteristics of the periodically oscillating T waves were well preserved after transformation, the tHR and rHR values were consistent, and the ptSQI values were high because of the absence of noise. For the acceptable ECGs, the ptSQI values were low, and the tHRs deviated from the rHRs because the frequency bands of the motion interference and T wave overlapped with each other. For the unacceptable ECGs, the ptSQI values were still low, and both the tHRs and rHRs deviated from the actual heart rate values because of the strong motion interference.

Table 2 lists the indices that were calculated from the segments in the training dataset with different quality levels. When compared with the good ECGs, the $adSQI_1$, $adSQI_2$, ptSQI, and tHR values were significantly lower,

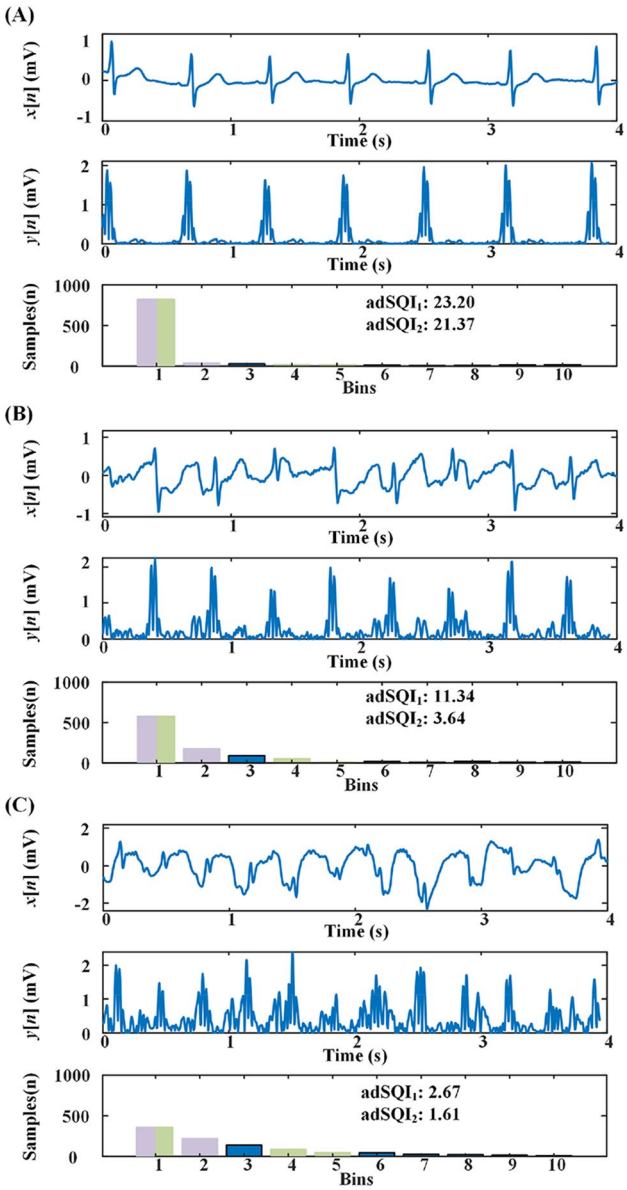

Fig. 4 The $adSQI_1$ and $adSQI_2$ indices at different quality levels. **A** Segment labeled as a good ECG; **B** segment labeled as an acceptable ECG; **C** segment labeled as an unacceptable ECG

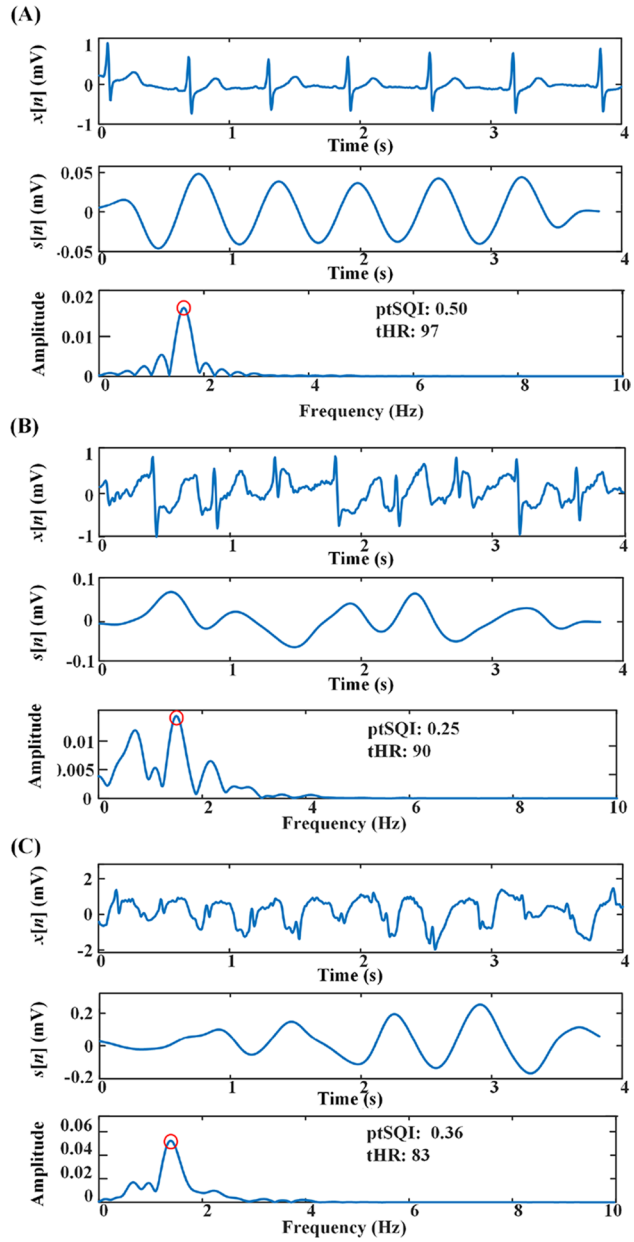

Fig. 5 The ptSQI and tHR indices at different quality levels. **A** Segment labeled as a good ECG; **B** segment labeled as an acceptable ECG; **C** segment labeled as an unacceptable ECG

while the rHRs were significantly higher for the acceptable ECGs and unacceptable ECGs. When compared with the acceptable ECGs, the $adSQI_1$, $adSQI_2$, ptSQI, and tHR values were considerably lower, while the rHRs were significantly higher for the unacceptable ECGs.

3.2 The performance of signal quality assessment

Table 3 provides the relevant coincidence matrix of the signal quality assessment results for both the training and testing datasets. The sensitivities for identifying unacceptable, acceptable, and good ECGs were 93.33%, 95.80%, and 97.40%, respectively, for the training dataset, and were 94.41%, 96.51%, and 97.79% for the testing dataset, respectively. The overall accuracy and kappa values were 96.11% and 0.94, respectively, for the training dataset and were 96.74% and 0.95 for the testing dataset, respectively. There were no significant differences in the sensitivity, accuracy, or kappa values between the training dataset and testing dataset.

Table 2 Indices used for signal quality assessments for different annotation levels in the training dataset

Index	Annotation label		
	Good	Acceptable	Unacceptable
$adSQI_1$	23.64 (20.17, 30.23)	19.05 (14.69, 24.66) *	5.04 (3.05, 7.37) *#
$adSQI_2$	23.04 (13.58, 30.90)	7.65 (4.66, 16.51) *	1.95 (1.61, 2.44) *#
ptSQI	0.47 (0.43, 0.51)	0.29 (0.25, 0.33) *	0.28 (0.24, 0.32) *#
tHR	92.47 (84.23, 102.54)	86.98 (65, 106.20) *	73.24 (59.51, 86.98) *#
rHR	92.47 (84.23, 102.54)	108.03 (95.21, 125.43) *	150.15 (120.85, 169.37) *#

*Compared with good ECG, $p < 0.05$; comparison among acceptable ECG, $p < 0.05$

Table 3 Performance of the proposed method for signal quality assessments

Annotation	SQG (Training dataset)					SQG (Testing dataset)				
	0	1	2	Sen	Kappa	0	1	2	Sen	Kappa
Unacceptable	740	52	1	93.33%	0.94	321	19	0	94.41%	
Acceptable	35	1918	49	95.80%		13	829	17	96.51%	0.95
Good	0	57	2133	97.40%		0	21	928	97.79%	
Acc	96.11%					96.74%				

SQG signal quality grading; *Sen* sensitivity; *Acc* accuracy

Table 4 Detection results for R waves and T waves

Feature	SQG					
Annotation	1			2		
R wave	Total	Correct	Acc(%)	Total	Correct	Acc(%)
Unacceptable	166	162	97.59	0	0	-
Acceptable	6272	6270	99.97	112	112	100.00
Good	145	145	100.00	5872	5872	100.00
T wave				Total	Correct	Acc(%)
Unacceptable		0	0	-		
Acceptable		105	85	80.95		
Good		5468	5464	99.93		

SQG signal quality grading; *Acc* accuracy

3.3 R wave and T wave detection results

Table 4 lists the R wave and T wave detection results for the segments that were assigned SQG values of 1 or 2 by the signal quality assessment algorithm. The overall accuracy was 99.95% for R wave detection. For the segments with SQG values of 1, the accuracies were 97.59%, 99.97%, and 100% when the ECGs were annotated as unacceptable, acceptable, and good, respectively. For the segments with SQG values of 2, all of the R waves were correctly detected either the ECGs were annotated as acceptable or annotated as good. The overall accuracy was 99.57% for T wave detection. For the segments that were annotated as acceptable but with SQGs of 2, 80.95% of the T waves were correctly detected. For the segments that were annotated as good and with SQGs of 2, the detection accuracy was 99.93%.

3.4 Simulation of real-time application

Figure 6 shows the simulation results of the real-time application of the proposed algorithm. In this case, the ECG and acceleration signals were continuously recorded during different physical activities of the volunteers, which included resting, walking, jogging, running, and creeping. The total recording time was 13.87 min, and a total of 208 consecutive segments were analyzed. One hundred eight, 90, and 10 segments were classified as good, acceptable, or unacceptable ECGs, respectively. The lower panel shows the dynamic heart rate, as calculated from the tHR, which is an index that was used for signal quality assessment with a time interval of 4 s. If the segment was classified as good or acceptable, the original value was displayed and labeled green or yellow. Otherwise, the heart rate was estimated as the average of the 3 previous consecutive values if the segment was classified as unacceptable. As shown in the lower panel, most

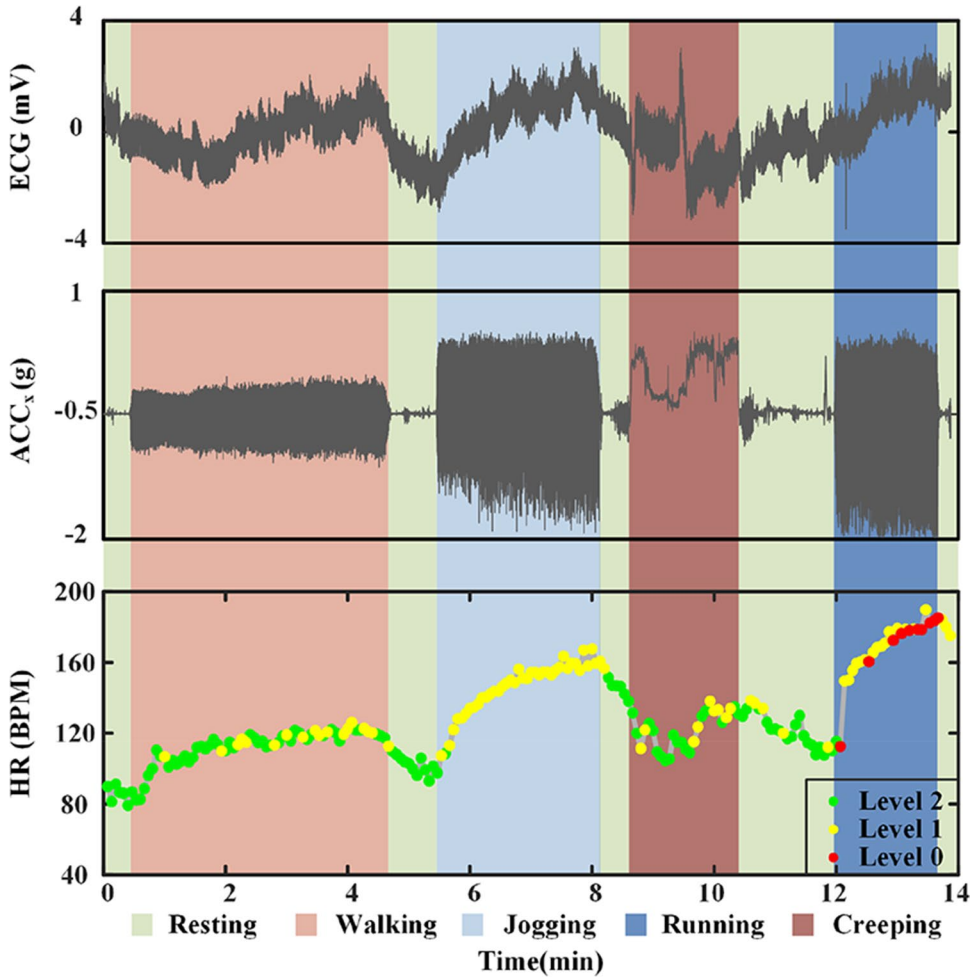

Fig. 6 A simulation of the real-time application of the proposed algorithm. ACC acceleration, HR heart rate, BMP beats per minute

segments were classified as good during resting, walking, and creeping. Conventional methods, such as R wave and T wave morphology analysis and HRV analysis, can be used to delineate the ECG waveforms. During jogging, R wave morphology analysis and HRV analysis can be used to delineate the ECG waveforms since most segments were classified as acceptable. During running, the ECG segments were classified as either unacceptable or acceptable, so only the heart rate could be monitored.

4 Discussion

In this paper, we proposed an SQA-based ECG waveform delineation method to be used for wearable systems. Our results demonstrated that the proposed method could make the best use of all of the collected ECG signals by assessing and grading the signals into 3 quality levels. The developed method can be used to reliably analyze ECG waveforms during normal physical activities such as resting, walking, jogging, and creeping.

Since the PhysioNet/CinC Challenge 2011, many SQA methods have been developed to interpret the quality of ECG waveforms. These methods usually consist of three major stages: preprocessing, feature extraction, and classification [40–45]. The purpose of preprocessing is to suppress the different type's artifacts that are present in ECG waveforms. The purpose of feature extraction is to extract the fiducial and/or the non-fiducial features of the ECG signals. Moody and Mark [40] proposed the first method to evaluate the noise content in ECG recordings. The ratio of the peak-to-peak magnitude to the estimated noise level of each lead was used to discard highly corrupted ECG leads before proceeding with beat classification. Li et al. [41] proposed a method for evaluating the physiological signal quality by fusing the kurtosis of the ECG (kSQI), comparing multiple beat detection algorithms on a single lead (bSQI), comparing beat detections using different ECG leads (iSQI) and the ECG spectrum distribution (sSQI) to perform SQA. Behar et al. [42] proposed two indices and termed them as the ratio of the sum of the eigenvalues associated with the five principal components over the sum of all eigenvalues that were obtained by principal component analysis (pcaSQI) and the

ratio of the number of beats detected by different algorithms (rSQI). By combining the two indices with the aforementioned indices, the classification accuracy reached 97.10% when using the data from the 2011 PhysioNet/CinC Challenge. Three additional indices, which are called the index for assessing the interval anomalies in RR interval time series with fixed time windows (iSQI), index for assessing the morphology consistency of any two ECG beats (tSQI), and index for assessing high-amplitude artifacts in ECG (aSQI), were combined with the aforementioned indices to access the quality of ECGs for QRS detection by Liu et al. [43]. By using long-term ECG data that were collected from a SmartVest system during daily activities, their method correctly identified 97.90% good-quality and 96.40% poor-quality ECGs. Castro et al. [44] presented a template correlation SQI (corrSQI) to classify usable signals from contactless capacitively coupled ECGs for the extraction of sleep apnea features with accuracies between 86 and 91%. Nardelli et al. [45] described a higher-order statistics SQI (hosSQI) for the real-time evaluations of ECG recording quality. By using data from the 2011 PhysioNet/CinC Challenge, an accuracy of 90.38% was achieved for signal quality discrimination. These studies, however, concentrated on features that are related to R waves while the features related to T waves from the signal morphology present a greater challenge.

The classification stage involves grading the ECG signals using heuristic rules or machine learning techniques. In published works, the ECG signal quality was categorized into two, three, or five quality levels [18, 33, 46–48]. The ECG signals were classified as either acceptable or unacceptable in earlier studies, which was based on whether the heart rate could be accurately calculated or whether the underlying cardiac rhythm could be accurately determined [47]. To compensate for the natural disadvantages of a binary classification, Redmond et al. [48] classified the ECG quality into three categories according to whether heart rate was easy to determine (good), difficult but possible to determine (average), or could not reliably be determined (bad). Unfortunately, this method still aimed at whether the heart rate could be extracted reliably. Recently, Smital et al. [18] also classified ECG signal qualities into three levels but with different criteria: signals suitable for full waveform analysis (Q1), signals suitable only for QRS detection (Q2), and signals not suitable for further processing (Q3). The proposed method has been shown to be helpful for subsequent ECG analysis using artificially created ECG datasets and a real dataset that was acquired by a specially designed wearable ECG device. In addition, Li et al. [33] proposed a five-level ECG signal quality classification based on noise interference and whether it would interfere with the interpretation or recognition of P, T, or atrial flutter waves, namely, clean, minor noise, moderate noise, severe noise, and extreme noise. However, the classification standards overlapped between different levels, and how to perform further analysis according to these classifications was still unclear.

In the present study, an SQA-based ECG waveform delineation method was proposed and tested using ECG data that were recorded from wearable systems during different physical activities. On the one hand, five features that are related to R waves and T waves were used to assess the signal quality. In addition to the R wave–related index, rHR, four novel indices were proposed to characterize the motion-related artifacts and T waves. The two amplitude distribution–related features, $adSQI_1$ and $adSQI_2$, are correlated with the overall SNR. The additional two T wave–related features, ptSQI and tHR, are associated with the degree of deformation of the T wave. The consistency levels between the rHRs and tHRs are correlated with the noise levels at different frequency bands. On the other hand, similar to Redmond et al. [48] and Smital et al. [18], three quality levels were used to categorize the signal quality. However, the annotation criteria were slightly different between the two studies. Good ECGs that could be used for full waveform analysis were characterized by clear R waves and T waves and by relatively steady lines between beats. Acceptable ECGs that were suitable for QRS detection and HRV analysis were characterized by identifiable R waves but distorted T waves. The high accuracies for the quality level classification, R wave detection, and T wave detection indicated that the proposed method could make the best use of the collected ECG signals for the early diagnosis and prevention of CVDs during normal physical activities.

The possible limitations should be discussed. First, the signal quality label is artificially defined, and there is no uniform and strict limit on the signal quality. The method that is used to assign the reference quality level may be somewhat subjective. Second, the algorithm was developed and tested on ECG waveforms that were recorded from healthy subjects. Therefore, the performance of this method still needs to be validated for subjects with CVDs. Third, the method was developed based on single-channel ECG analysis, without considering whether the interchannel agreement indices can benefit the SQA. Fourth, this quality assessment was only related to the determinations of the R peak and T peak locations exclusively and not to the full delineation of the full waveform. For example, this method did not provide any information regarding the possibility of accurately calculating the QT intervals.

5 Conclusion

In this paper, an SQA-based ECG waveform delineation method that was used for wearable systems was developed. Five indices, including the multiscale nonlinear amplitude statistical distribution, the proportion of energy-related to

T waves, and heart rates calculated from R waves and T waves were used to categorize the signal quality. The signals were classified as good, acceptable, or unacceptable ECGs by combining these indices using a neural network. The R waves and/or T waves were identified for the corresponding feature interpretations based on the SQA results. By using a real ECG waveform that was recorded from a wearable system, the SQA-based ECG waveform delineation method was shown to be capable of performing a reliable analysis and has the potential to be applied in wearable systems for the early diagnosis and prevention of CVDs.

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Declarations

Conflict of interest The authors declare that there are no conflicts of interest.

References

1. McGill HC Jr, McMahan CA, Gidding SS (2008) Preventing heart disease in the 21st century: implications of the Pathobiological Determinants of Atherosclerosis in Youth (PDAY) study. Circulation 117(9):1216–1227
2. Jain SK, Bhaumik B (2017) An energy efficient ECG signal processor detecting cardiovascular diseases on smartphone. IEEE Trans Biomed Circuits Syst 11(2):314–323
3. Köhler BU, Hennig C, Orglmeister R (2002) The principles of software QRS detection. IEEE Eng Med Biol Mag 21(1):42–57
4. Alaei S, Wang S, Anaya P, Patwardhan A (2020) Co-occurrence and phase relationship between alternans of the R wave amplitude (RWAA) and of the T wave (TWA) in ECGs. Comput Biol Med 121:103785
5. El-Battrawy I, Lan H, Cyganek L, Zhao Z, Li X, Buljubasic F, Lang S, Yücel G, Sattler K, Zimmermann WH, Utikal J, Wieland T, Ravens U, Borggrefe M, Zhou XB, Akin I (2018) Modeling short QT syndrome using human-induced pluripotent stem cell-derived cardiomyocytes. J Am Heart Assoc 7(7):e007394
6. Rajendra Acharya U, Paul Joseph K, Kannathal N, Lim CM, Suri JS (2006) Heart rate variability: a review. Med Biol EngComput 44(12):1031–1051
7. Feeny AK, Chung MK, Madabhushi A, Attia ZI, Cikes M, Firouznia M et al (2020) Artificial intelligence and machine learning in arrhythmias and cardiac electrophysiology. Circ Arrhythm Electrophysiol 13(8):e007952
8. Drew BJ, Califf RM, Funk M, Kaufman ES, Krucoff MW, Laks MM, et al (2004) American Heart Association; Councils on Cardiovascular Nursing, Clinical Cardiology, and Cardiovascular Disease in the Young. Practice standards for electrocardiographic monitoring in hospital settings: an American Heart Association scientific statement from the Councils on Cardiovascular Nursing, Clinical Cardiology, and Cardiovascular Disease in the Young: endorsed by the International Society of Computerized Electrocardiology and the American Association of Critical-Care Nurses. Circulation 110(17):2721–2746.
9. Attia ZI, Noseworthy PA, Lopez-Jimenez F, Asirvatham SJ, Deshmukh AJ, Gersh BJ et al (2019) An artificial intelligence-enabled ECG algorithm for the identification of patients with atrial fibrillation during sinus rhythm: a retrospective analysis of outcome prediction. Lancet 394(10201):861–867
10. Laferriere P, Chan ADC, Lemaire ED (2011) Surface electromyographic signals using dry electrodes. IEEE Trans Instrum Meas 60(10):3259–3268
11. Miao F, Cheng Y, He Y, He Q, Li Y (2015) A wearable context-aware ECG monitoring system integrated with built-in kinematic sensors of the smartphone. Sensors (Basel) 15(5):11465–114684
12. Huang S, Li J, Zhang P, Zhang W (2018) Detection of mental fatigue state with wearable ECG devices. Int J Med Inform 119:39–46
13. Koshy AN, Sajeev JK, Nerlekar N, Brown AJ, Rajakariar K, Zureik M et al (2018) Smart watches for heart rate assessment in atrial arrhythmias. Int J Cardiol 266:124–127
14. Li M, Xiong W, Li Y (2020) Wearable measurement of ECG signals based on smart clothing. Int J Telemed Appl 2020:6329360
15. Tian Y, Abdizadeh M, Mahnam A, Bhattachan P, Meghrazi MA et al (2020) Modeling and reconstructing textile sensor noise: implications for wearable technology. Annu Int Conf IEEE Eng Med Biol Soc 2020:4563–4566
16. Satija U, Ramkumar B, Manikandan MS (2018) Automated ECG noise detection and classification system for unsupervised healthcare monitoring. IEEE J Biomed Health Inform 22(3):722–732
17. Moeyersons J, Smets E, Morales J, Villa A, De Raedt W, Testelmans D et al (2019) Artefact detection and quality assessment of ambulatory ECG signals. Comput Methods Programs Biomed 182:105050
18. Smital L, Haider CR, Vitek M, Leinveber P, Jurak P, Nemcova A et al (2020) Real-time quality assessment of long-term ECG signals recorded by wearables in free-living conditions. IEEE Trans Biomed Eng 67(10):2721–2734
19. Di Marco LY, Chiari L (2011) A wavelet-based ECG delineation algorithm for 32-bit integer online processing. Biomed Eng Online 10:23
20. Berset T, Geng D, Romero I (2012) An optimized DSP implementation of adaptive filtering and ICA for motion artifact reduction in ambulatory ECG monitoring. Annu Int Conf IEEE Eng Med Biol Soc 2012:6496–6499
21. Wang Z, Wan F, Wong CM, Zhang L (2016) Adaptive Fourier decomposition based ECG denoising. Comput Biol Med 77:195–205
22. Paul JS, Reddy MR, Kumar VJ (2000) A transform domain SVD filter for suppression of muscle noise artefacts in exercise ECG's. IEEE Trans Biomed Eng 47(5):654–663
23. Chen M, Zhang X, Chen X, Zhu M, Li G, Zhou P (2016) FastICA peel-off for ECG interference removal from surface EMG. Biomed Eng Online 15(1):65
24. Jekova I, Krasteva V, Christov I, Abächerli R (2012) Threshold-based system for noise detection in multilead ECG recordings. Physiol Meas 33(9):1463–1477
25. Castiglioni P, Meriggi P, Faini A, Rienzo MD (2011) Cepstral based approach for online quantification of ECG quality in freely moving subjects. Computing in Cardiology 38:625–628
26. Xia H, Garcia GA, Bains J, Wortham DC, Zhao X (2012) Matrix of regularity for improving the quality of ECGs. Physiol Meas 33(9):1535–1548
27. Satija U, Ramkumar B, Manikandan MS (2017) Real-time signal quality-aware ECG telemetry system for IoT-based health care monitoring. IEEE Internet Things J 4(3):815–823
28. Satija U, Ramkumar B, Manikandan MS (2018) A review of signal processing techniques for electrocardiogram signal quality assessment. IEEE Rev Biomed Eng 11:36–52

29. Ong ME, Lee Ng CH, Goh K, Liu N, Koh ZX, Shahidah N et al (2012) Prediction of cardiac arrest in critically ill patients presenting to the emergency department using a machine learning score incorporating heart rate variability compared with the modified early warning score. Crit Care 16(3):R108
30. Verrier RL, Ikeda T (2013) Ambulatory ECG-based T-wave alternans monitoring for risk assessment and guiding medical therapy: mechanisms and clinical applications. Prog Cardiovasc Dis 56(2):172–185
31. Holkeri A, Eranti A, Haukilahti MAE, Kerola T, Kenttä TV, Tikkanen JT et al (2020) Predicting sudden cardiac death in a general population using an electrocardiographic risk score. Heart 106(6):427–433
32. Abdul-Kadir NA, Mat Safri N, Othman MA (2016) Dynamic ECG features for atrial fibrillation recognition. Comput Methods Programs Biomed 136:143–150
33. Li Q, Rajagopalan C, Clifford GD (2014) A machine learning approach to multi-level ECG signal quality classification. Comput Methods Programs Biomed 117(3):435–447
34. Clifford GD, Behar J, Li Q, Rezek I (2012) Signal quality indices and data fusion for determining clinical acceptability of electrocardiograms. Physiol Meas 33(9):1419–1433
35. Huerta A, Martínez-Rodrigo A, González V, Quesada A, Rieta J, et al (2019) Quality assessment of very long-term ECG recordings using a convolutional neural network. 2019 E-Health and Bioengineering Conference (EHB). IEEE, pp 1–4.
36. Behar J, Johnson A, Clifford GD, Oster J (2014) A comparison of single channel fetal ECG extraction methods. Ann Biomed Eng 42(6):1340–1353
37. Behar J, Oster J, Clifford GD (2014) Combining and benchmarking methods of foetal ECG extraction without maternal or scalp electrode data. Physiol Meas 35(8):1569–1589
38. Johnson AE, Behar J, Andreotti F, Clifford GD, Oster J (2015) Multimodal heart beat detection using signal quality indices. Physiol Meas 36(8):1665–1677
39. Ranjith P, Baby P, Joseph P (2003) ECG analysis using wavelet transform: application to myocardial ischemia detection. ITBM-RBM 24(1):44–47
40. Moody GB, Mark RG (1982) Development and evaluation of a 2-lead ecg analysis program. Comput Cardiol 9:39–44
41. Li Q, Mark RG, Clifford GD (2008) Robust heart rate estimation from multiple asynchronous noisy sources using signal quality indices and a Kalman filter. Physiol Meas 29(1):15–32
42. Behar J, Oster J, Li Q, Clifford GD (2013) ECG signal quality during arrhythmia and its application to false alarm reduction. IEEE Trans Biomed Eng 60(6):1660–1666
43. Liu C, Zhang X, Zhao L, Liu F, Chen X, Yao Y, Li J (2018) Signal quality assessment and lightweight QRS detection for wearable ECG SmartVest system. IEEE Internet Things J 6:1363–1374
44. Castro ID, Varon C, Torfs T, Van Huffel S, Puers R, Van Hoof C (2018) Evaluation of a multichannel non-contact ECG system and signal quality algorithms for sleep apnea detection and monitoring. Sensors (Basel) 18(2):577
45. Nardelli M, Ianata A, Valenza G, Felicib M, Baraglib P, Scilingoa EP (2020) A tool for the real-time evaluation of ECG signal quality and activity: application to submaximal treadmill test in horses. Biomedical Signal Processing and Control 56:101666
46. Hayn D, Jammerbund B, Schreier G (2011) ECG quality assessment for patient empowerment in mHealth applications. Computing in Cardiology 38:353–356
47. Orphanidou C, Bonnici T, Charlton P, Clifton D, Vallance D, Tarassenko L (2015) Signal-quality indices for the electrocardiogram and photoplethysmogram: derivation and applications to wireless monitoring. IEEE J Biomed Health Inform 19(3):832–838
48. Redmond SJ, Xie Y, Chang D, Basilakis J, Lovell NH (2012) Electrocardiogram signal quality measures for unsupervised telehealth environments. Physiol Meas 33(9):1517–1533

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